Confirmatory Factor Analysis

PSYC 7304 / 8304 · Measurement Theory & Psychometrics · Spring 2026

Confirmatory Factor Analysis

Model specification · Estimation · Fit evaluation · Reliability & replicability

EFA vs. CFA: The Key Shift

Exploratory Factor Analysis

- No prior structure imposed
- Every indicator free to load on every factor
- Data determine the solution
- Goal: *discover* dimensionality
- Appropriate early in scale development

Confirmatory Factor Analysis

- Structure specified *a priori* from theory
- Non-hypothesized cross-loadings **constrained to zero**
- Model fit against data is evaluated
- Goal: *test* a theoretically derived structure
- Appropriate when you have a hypothesis to evaluate

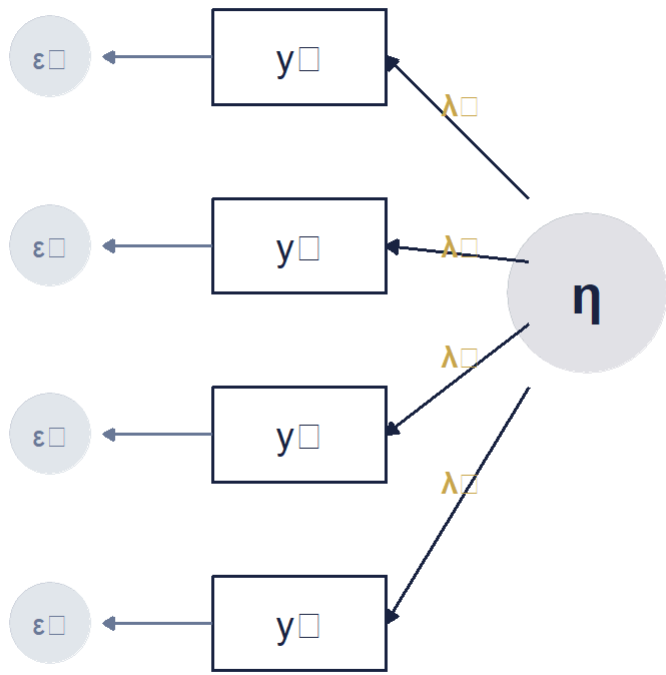
The zero constraint on unspecified loadings is CFA's core feature — and the source of both its inferential power and its fit sensitivity. Every zero you impose is a testable claim about the data.

The CFA Model

For each observed indicator y_i:

$$y_i = \lambda_i \eta + \varepsilon_i$$

Parameter	Symbol	Meaning
Factor loading	λ	Regression of indicator on factor
Factor variance	Φ	Variance (and covariance) of latent variable(s)
Residual variance	θ	Unique variance — error + item specificity



Identification — the latent variable has no natural scale. Fix it one of two ways:

- **Marker variable**: fix one loading per factor to 1.0
- **Standardization**: fix factor variance to 1.0

Degrees of freedom = $p(p+1)/2 - q$
where p = indicators, q = free parameters

Estimation

Maximum Likelihood (ML) — the default

Finds parameter values that maximize the likelihood of observing the sample covariance matrix given the model.

Assumes: - Multivariate normality - Continuous indicators - Adequate sample size (N ≥ ~200 as a rough floor)

When ML assumptions fail:

Situation	Estimator	Notes
Non-normal continuous data	**MLR**	Satorra-Bentler scaled χ^2
Ordinal indicators (≥5 categories)	**WLSMV**	Polychoric correlations
Ordinal indicators (<5 categories)	**WLSMV**	Treat as categorical
Missing data	**FIML with ML or MLR**	Full information ML

> Violating ML assumptions inflates χ^2 and distorts fit indices. Using ML on Likert data without checking is one of the most common errors in the applied CFA literature.

The CFA Procedure

#	Stage	Action	Caution
1	**Specify**	Define indicators per factor; constrain non-hypothesized loadings to zero	
2	**Identify**	Confirm df > 0; set factor scale (marker variable or standardize)	
3	**Estimate**	Run ML (or MLR / WLSMV); examine convergence and Heywood cases	⚠ *Non-convergence = identification problem*
4	**Evaluate fit**	Examine global fit (χ^2, RMSEA, CFI, TLI, SRMR) and local fit (residuals, MIs)	⚠ *No single index is sufficient*
5	**Modify**	If fit is poor: inspect MIs and residuals for substantive misspecification	⚠ *MI-driven changes are exploratory — capitalize on chance*
6	**Replicate**	Treat any modified model as a new hypothesis requiring independent replication	⚠ *Modified models must not be reported as confirmatory*

Evaluating Model Fit

Global fit · Absolute indices · Relative indices · Local fit

The Chi-Square Test

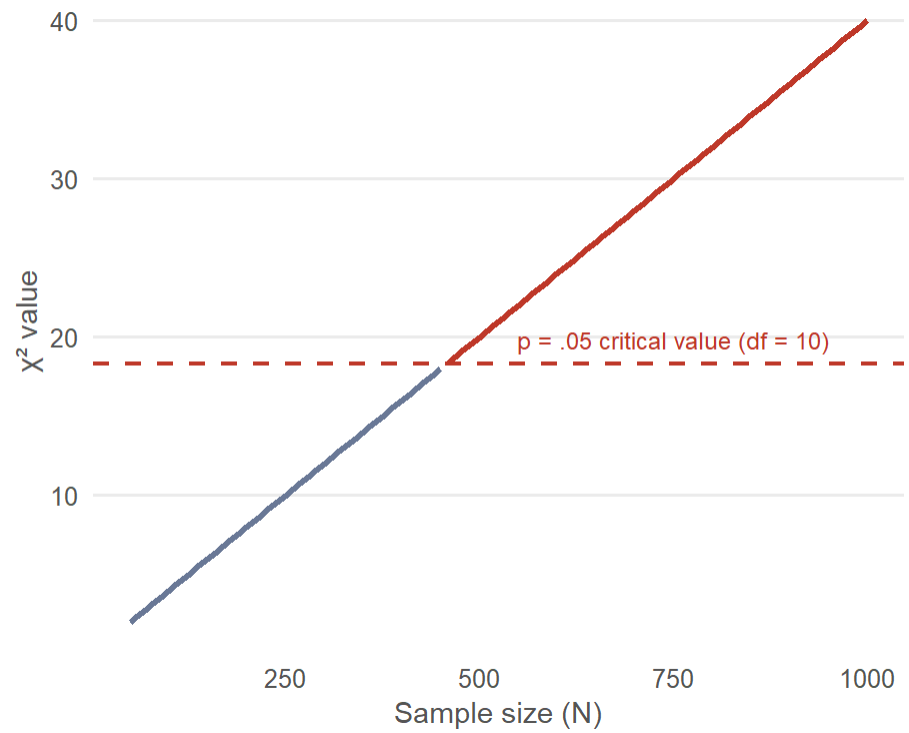
H_0: The model-implied covariance matrix equals the population covariance matrix (exact fit)

$$\chi^2 = (N - 1) \cdot F_{ML}$$

Problems with χ^2 alone:

- With large *N*: detects trivially small, practically irrelevant misfit → almost always significant
- With small *N*: underpowered to detect real misfit → falsely retained models
- Tests *exact* fit — a standard no structural model ever truly meets

> Never use χ^2 as the sole fit criterion. Always report it — it is the foundation of approximate fit indices — but interpret in context.



Same model misfit (F_{ML} = 0.04), only *N* varies. χ^2 crosses significance around *N* = 450.

Absolute Fit Indices

Assess how well the model reproduces the observed covariances — **no reference to a baseline model**

RMSEA — Root Mean Square Error of Approximation

$$\text{RMSEA} = \sqrt{\frac{\chi^2 - df}{df(N-1)}}$$

Penalizes model complexity (larger *df* → smaller RMSEA for same misfit). Always report the **90% CI**.

Value	Interpretation
≤ .05	Close fit
.05–.08	Reasonable
.08–.10	Mediocre
> .10	Poor

RMSEA is biased against simple models with few df and small N. Do not over-rely on it for models with fewer than ~12 indicators.

SRMR — Standardized Root Mean Square Residual

$$\text{SRMR} = \sqrt{\frac{2\sum_{i \geq j}(s_{ij} - \hat{\sigma}_{ij})^2}{p(p+1)}}$$

Average discrepancy between observed and model-implied correlations. Directly interpretable as a mean residual correlation.

Value	Interpretation
≤ .05	Excellent
≤ .08	Acceptable
> .10	Poor

SRMR is especially sensitive to misspecified **factor loadings**. If SRMR is large but RMSEA is acceptable, suspect loading misspecification rather than structural misfit.

Relative (Incremental) Fit Indices

Compare the target model to a **null model** (all indicators uncorrelated — the worst-case baseline)

CFI — Comparative Fit Index

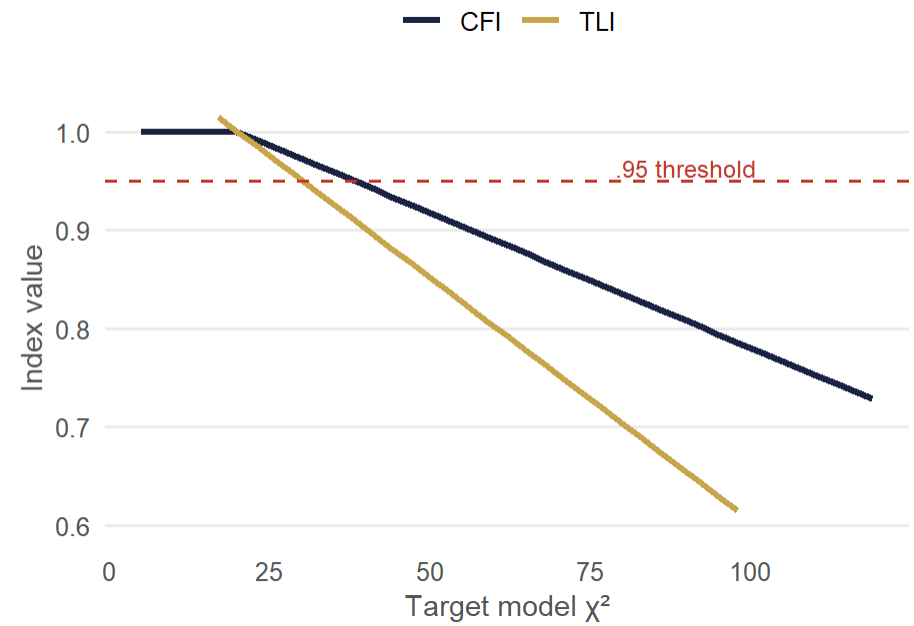
$$\mathrm{CFI} = 1 - \frac{\max(\chi_t^2 - df_t,\ 0)}{\max(\chi_0^2 - df_0,\ 0)}$$

- Ranges 0–1 (values > 1 truncated to 1)
- Relatively robust to sample size
- Does **not** penalize model complexity

TLI — Tucker-Lewis Index

$$\mathrm{TLI} = \frac{\chi_0^2/df_0 - \chi_t^2/df_t}{\chi_0^2/df_0 - 1}$$

- Penalizes for model complexity — parsimonious models get a bonus
- Can exceed 1.0; round to 1.0
- Prefer TLI when comparing models of different complexity



Index	Hu & Bentler threshold
CFI	≥ .95 (≥ .90 older literature)
TLI	≥ .95

The Hu & Bentler Problem

Why the .95/.06/.08 cutoffs are heuristics, not hard criteria

The widely-cited cutoffs come from **one simulation study** (Hu & Bentler, 1999) conducted under specific conditions:

- Continuous indicators, ML estimation
- N = 250
- Simple structure (3–4 factors, ~4 indicators each)
- Correctly specified models

What the cutoffs do not account for:

- Small N → RMSEA inflated, CFI unstable
- Ordinal data / WLSMV → different distributions
- Simple models (few df) → RMSEA biased upward
- Complex models → CFI biased downward
- Model misspecification type matters

Marsh, Hau, & Wen (2004); McCallum et al. (2006); and others have shown the cutoffs are sample-size and complexity dependent. A CFI of .93 in a complex model at N = 150 may indicate better fit than CFI = .96 in a simple model at N = 500.

Current best practice: Report all indices. Interpret the *pattern*. Use cutoffs as soft heuristics, not binary pass/fail criteria. The field is moving toward equivalence-testing approaches (SESOI-based fit) but these are not yet standard.

Information Criteria: AIC & BIC

For **model comparison** — not absolute fit assessment

$$\text{AIC} = \chi^2 - 2 \cdot df$$

$$\text{BIC} = \chi^2 - \ln(N) \cdot df$$

- Lower = better (more parsimonious fit)
- No absolute threshold — differences between models are what matter
- AIC penalizes less → rewards fit
- BIC penalizes more → rewards parsimony

ΔBIC interpretation (rough heuristics):

ΔBIC	Evidence for better model
0–2	Weak
2–6	Positive
6–10	Strong
> 10	Very strong

Example: *Comparing a 1-factor vs. 2-factor model of a 10-item scale (N = 300)*

Model	*χ^2(df)*	*CFI*	*RMSEA*	*AIC*	*BIC*
1-factor	*87.4 (35)*	*.91*	*.070*	*3,421*	*3,498*
2-factor	*44.2 (34)*	*.98*	*.033*	*3,380*	*3,464*

ΔBIC = 34 → very strong evidence for 2-factor model

$\Delta\chi^2(1) = 43.2$, $p < .001$ → significant improvement

Also supported by CFI, RMSEA pattern

Use AIC/BIC when comparing **nested or non-nested alternative models** — not as a substitute for absolute fit evaluation.

Local Fit: Residuals & Modification Indices

After evaluating global fit — **where specifically** does the model misfit?

Residual correlations

Difference between observed and model-implied correlation for each indicator pair.

- |residual| > .10 → the model is not explaining this pair well
- Cluster of large residuals among specific items → may indicate a method factor or missing subfactor

Modification indices (MI)

Estimated $\Delta\chi^2$ if a currently-constrained parameter were freed.

- MI > 10 → investigate
- Large MI on a cross-loading → that item may belong to (or share variance with) another factor
- Large MI on a residual covariance → those two items share specific variance the model doesn't capture (e.g., similar wording, same reverse-scored)

The specification search problem

Acting on MIs without substantive justification produces overfit models. Each freed parameter:

- Reduces df
- Improves fit in *this* sample
- Capitalizes on chance
- Will not replicate

The defensible approach: 1. Is there a *substantive* reason this parameter should be non-zero? 2. Was it overlooked in the original theory, or is it a genuine surprise? 3. If freed: label the model as revised/exploratory 4. Require replication in an independent sample

Reliability & Factor Replicability

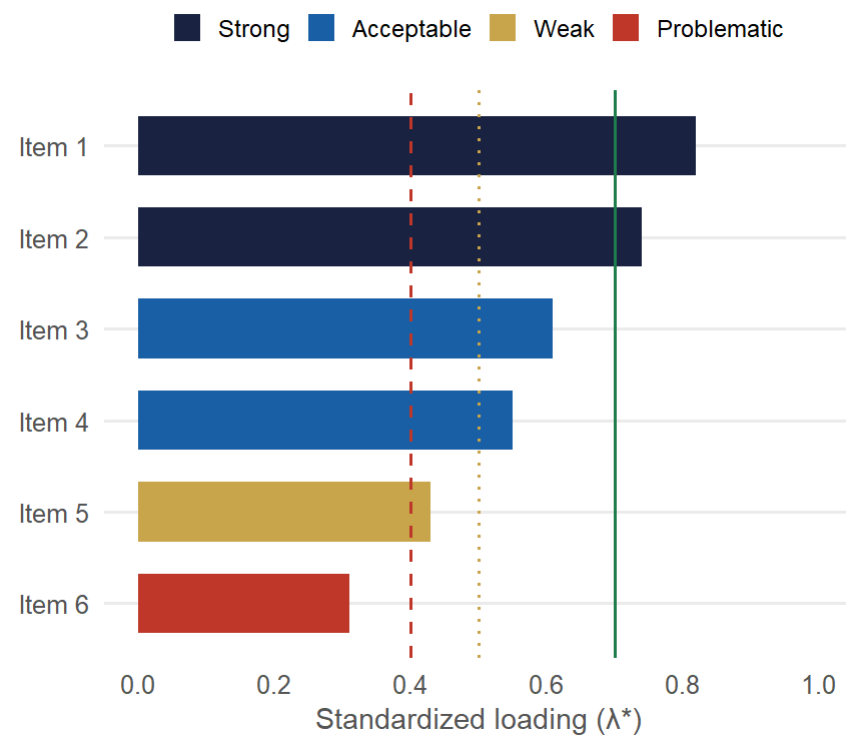
Model-based reliability · ω and ω_h · H · AVE · Factor determinacy

Factor Loadings & What They Tell You

Standardized loadings (λ*) are correlations between the indicator and the latent factor.

λ*	Interpretation
≥ .70	Strong — indicator is a good marker
.50–.70	Acceptable
.40–.50	Weak but interpretable
< .40	Problematic — consider revising or dropping

Heywood cases — standardized loading > 1.0 or residual variance < 0 — indicate identification problems, near-multicollinearity, or too few indicators per factor. Not a fit index issue; a model specification issue.



Model-Based Reliability: ω and ω_h

McDonald's ω (omega)

$$\omega = \frac{(\sum \lambda_i)^2}{(\sum \lambda_i)^2 + \sum \theta_i}$$

Proportion of total score variance due to the common factor. Does **not** assume tau-equivalence (equal loadings) — unlike α.

> α is a special case of ω when all loadings are equal. When loadings differ (almost always), α and ω diverge. Prefer ω.

α vs. ω comparison (simulated 6-item scale):

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α = 0.744   ω = 0.758
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ω_h (omega hierarchical)

Relevant for multidimensional scales where a *total score* is used.

$$\omega_h = \frac{(\sum \lambda_g)^2}{(\sum \lambda_g)^2 + \sum \lambda_s^2 + \sum \theta_i}$$

where λ_g = general factor loadings, λ_s = specific factor loadings (from a bifactor model).

Interpretation:

- ω_h = proportion of total score variance attributable to the *general factor only*
- High ω + low ω_h → the scale is reliable, but the total score is not unidimensional
- If ω_h < .50, interpret subscale scores rather than the total

Example: A 20-item well-being scale has ω = .91, ω_h = .48. The scale is internally consistent, but total scores reflect multiple specific factors nearly as much as the general factor. Total scores may obscure meaningful subscale differences.

Factor Replicability: H and AVE

H — Construct Replicability Index (Hancock & Mueller, 2001)

$$H = \frac{1}{1 + \frac{1}{\sum_i \frac{\lambda_i^{*2}}{1 - \lambda_i^{*2}}}}$$

Probability that the latent factor will be recovered similarly across studies. Higher H = more stable, replicable factor.

H	Interpretation
≥ .80	Well-defined, likely to replicate
.70–.80	Acceptable
< .70	Poorly defined — factor may be unstable

H is sensitive to both the **number of indicators** and their **loadings**. A factor with 3 strong indicators ($\lambda^* \geq .70$) outperforms one with 6 weak indicators ($\lambda^* \approx .35$).

AVE — Average Variance Extracted (Fornell & Larcker, 1981)

$$\text{AVE} = \frac{\sum \lambda_i^{*2}}{n}$$

Average communality across indicators — the proportion of indicator variance the factor captures.

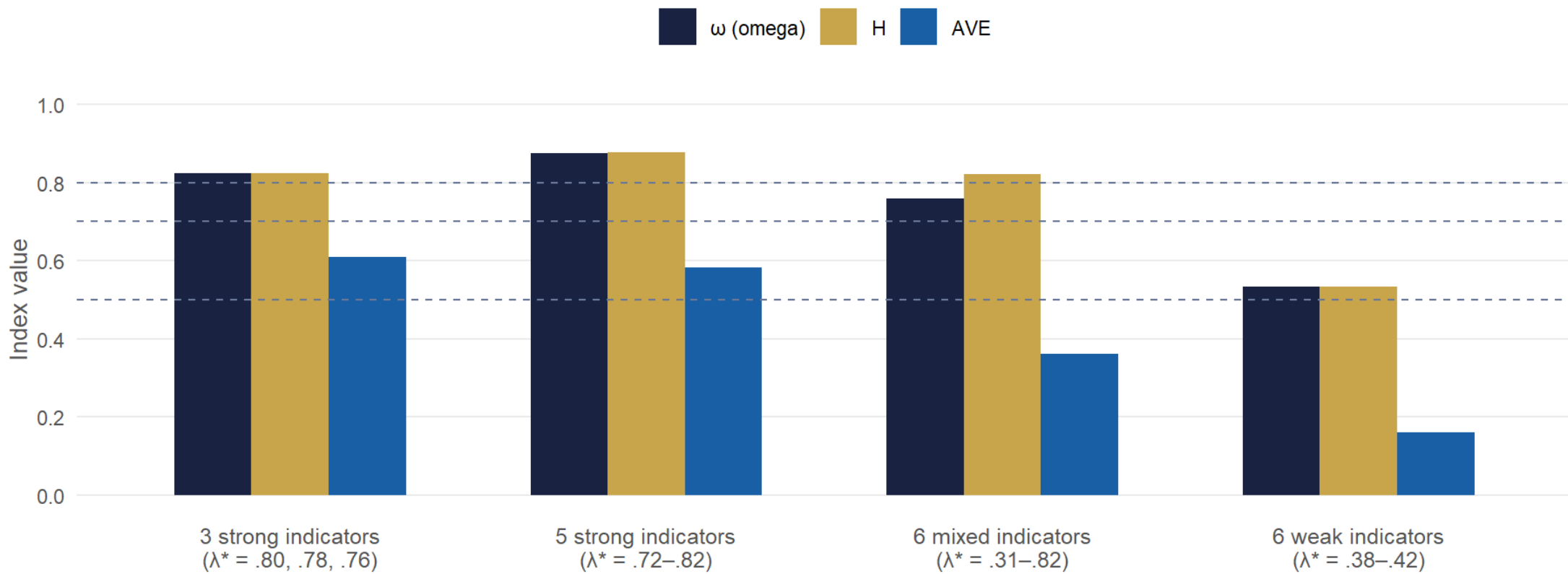
AVE	Interpretation
≥ .50	Factor explains more variance than error
< .50	Error variance dominates

Discriminant validity via Fornell-Larcker:

AVE for each factor should exceed the *squared* inter-factor correlation. If $\varphi^2_{12} > \text{AVE}_1$ **or** AVE_2, the two factors share more with each other than with their own indicators — discriminant validity is questionable.

Factor Replicability: H vs. ω

They answer different questions — both matter



Dashed lines at .50 (AVE threshold), .70 (H minimum), and .80 (H preferred). Note: 6 weak indicators can produce acceptable ω but poor H — the scale is consistent but the factor won't replicate.

Factor Determinacy (FD)

How well can we estimate individuals' latent factor scores?

Factor determinacy is the correlation between the true latent factor scores and the estimated factor scores (the weighted composites you'd save for use in subsequent analyses):

$$FD = \sqrt{\lambda' \Sigma^{-1} \lambda}$$

where Σ is the model-implied indicator covariance matrix.

FD	Interpretation
≥ .90	Factor scores are good proxies for the latent variable
.80–.90	Acceptable with caution
< .80	Factor score estimates are poor — avoid using them as observed variables

Low FD means saving factor scores and treating them as observed variables in subsequent analyses introduces substantial error. Use latent variable SEM instead.

Example: *A researcher runs CFA on a 4-item stress scale (all loadings ≈ .45) and saves the factor scores to use as a predictor in a regression.*

N = 180, FD = .74

The factor score estimates correlate only .74 with the true latent variable. Using them as a predictor in regression attenuates regression coefficients — the same problem as using an unreliable observed score.

The fix: keep the latent variable in the model and run SEM rather than a two-step regression on factor scores.

Putting It Together

The full reporting checklist

What to Report

Minimum required reporting:

Element	Report as
Sample & estimator	N, estimator (ML / MLR / WLSMV), software
Chi-square	$\chi^2(df)$, p
RMSEA	RMSEA, 90% CI
CFI	CFI
TLI	TLI
SRMR	SRMR
Factor loadings	Standardized λ* with SE or 95% CI
Reliability	ω per factor (not α alone)
Replicability	H and AVE per factor

For model comparisons, add:

- $\Delta\chi^2(\Delta df)$ for nested models
- ΔCFI (meaningful change: > .01)
- ΔRMSEA (meaningful change: > .015)
- ΔBIC for non-nested models

A complete fit sentence looks like:

"Model fit was acceptable: $\chi^2(43) = 61.2$, $p = .031$; RMSEA = .048 [90% CI: .018, .072]; CFI = .963; TLI = .951; SRMR = .052. Standardized loadings ranged from .54 to .81. Factor reliability was ω = .87; H = .83; AVE = .52."

Fit indices tell you about the model. Loadings, ω, H, and AVE tell you about the *factor*. Report both.

Putting It All Together: A Worked Example

Scenario: CFA of a 9-item Resilience scale, 3 factors (Personal Strength, Social Support, Adaptability), 3 items each. *N* = 320, ML estimation.

Index	Value	Interpretation
χ^2(df)	34.1 (24)	*p = .080 — acceptable*
RMSEA [90% CI]	.038 [.000, .062]	*Close fit*
CFI	.971	*Excellent*
TLI	.958	*Excellent*
SRMR	.044	*Excellent*
ω (Personal Str.)	.84	*Good*
ω (Social Supp.)	.79	*Good*
ω (Adaptability)	.76	*Acceptable*
H (Personal Str.)	.85	*Well-defined*
H (Social Supp.)	.80	*Well-defined*
H (Adaptability)	.77	*Acceptable*
AVE (Personal Str.)	.58	*AVE ≥ .50 ✓*
AVE (Social Supp.)	.51	*AVE ≥ .50 ✓*
AVE (Adaptability)	.49	*Just below threshold — examine loadings*